

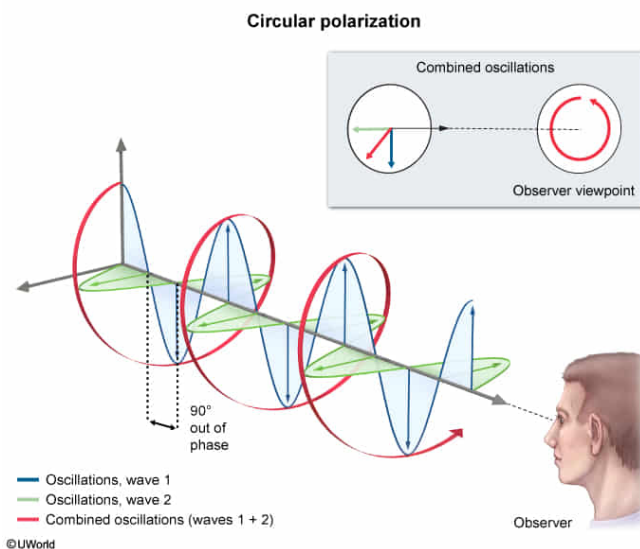
Circularly polarized light results from the vector sum of the electric field oscillations of two linearly polarized transverse waves traveling along the same axis. To achieve circular polarization, the two transverse waves must be out of phase to one another. The necessary phase difference is:

- ☐ A. 0° [1%]
- ✓ ☒ B. 90° [41%]
- ☐ C. 180° [54%]
- ☐ D. 360° [1%]

Correct

42% answered correctly

Explanation:



Polarization is a property unique to **transverse waves** (eg, electromagnetic radiation) in which the wave oscillations are aligned in a particular spatial orientation within the **x,y,z-coordinate system**. When two polarized waves combine, a new wave is formed with a different polarization. For example, a wave polarized along the x-axis totally **in-phase** with a wave polarized along the y-axis produces a **combined wave** that is polarized between the x- and y-axes at a 45° angle.

When the polarization of a combined wave **rotates** around the axis of propagation, the wave is said to be circularly polarized. If the polarization of the combined wave rotates in a clockwise direction when viewed facing the direction of propagation, the waveform is right-polarized; if the polarization rotates in a counterclockwise direction, the waveform is left-polarized.

Circular polarization occurs when two wave forms with **equal amplitude** and **perpendicular linear polarization** (ie, polarized along different axes) propagate **90° out of phase** to one another (ie, one waveform yields zero displacement when the other waveform is yielding a peak or a trough). The

mismatch in phase causes each contributing wave to generate momentary oscillations that are different at every point in time. This results in a continuous rotation in the orientation of the combined wave.

A 90° phase difference is required for circular polarization because only a 90° phase shift yields both the positive and negative oscillations along each axis and the symmetry necessary to produce a circular polarization pathway.

(Choices A and D) Phase differences of 0° and 360° both reflect waveforms that are traveling completely in phase with one another. The polarization of two waves traveling in phase with one another is fixed and will not rotate.

(Choice C) A phase shift of 180° represents waves traveling completely out of phase and will result in a polarization oriented in a fixed orientation at a 45° angle to the axis of polarization of each contributing wave.

Educational objective:

Circular polarization refers to the rotation of polarization observed when two transverse waves with equal amplitude and perpendicular linear polarization propagate out of phase with one another by 90°.

Time spent:0 sec

Subject:

Physics

QID:402431

System:

Light and Sound